Assignment # 2

Indvidual assignment

Chapter 1 : Introducing Models

Expectations:

Complete reading the entire chapter and then; in a similar manner to the predictive model which presented in chapter 1 of the HTMI book, submit a technical report to describe the overall modeling solution process to solve the attached **External Problem**.

1. In the submitted technical report also demonstrate the followings:
   a) Make a list of three useful **heuristics** found in this chapter.
   b) Mention at least three lessons you have learned from this chapter.
   c) Define the following terminologies:
      - Model
      - Trade-Offs
      - Accuracy
      - Variables and parameters
      - Robustness
      - Critique of Model
      - Sensitivity

External Problem

You are the Chief Engineer for a renewable-energy company operating two power-generation divisions: Division (S), a utility-scale Solar Farm, and Division (W), an onshore Wind Farm. The company has a specific capital budget for a strategic expansion plan that can only be applied to ONE of these two divisions. The investment increases the chosen division's operational capacity and monthly energy sales.

Characteristic	Division (S) – Solar Farm	Division (W) – Wind Farm
Profit Cycle	Long-term (PPA billed quarterly)	Short-term (sold daily/weekly on spot market)
Profit Margin	High: 35% – 55% of Monthly Sales	Low: 12% – 22% of Monthly Sales
Max. Monthly Sales	SR 150,000 – SR 320,000 (limited by daylight hours)	SR 280,000 – SR 650,000 (generates day and night)

Task I: Initial Predictive Modeling

Establish a predictive model to determine which division should receive the expansion investment to maximize total benefit over a two-year period.

a) Use a random generator to estimate model parameters within the ranges given above (margins and sales).

b) For your established model, evaluate the Robustness and Sensitivity of the results.

Task II: Impact of Technological Improvement

Assume the new investment includes a smart-grid predictive-maintenance software package that affects the divisions differently:

- Division (S): the profit range remains unchanged.
- Division (W): the software reduces turbine downtime and energy losses, increasing the profit margin by 12 percentage points (example: 12% becomes 24%).

a) Based on this new data, which division should receive the investment?

b) Evaluate the Robustness and Sensitivity of the results for this specific case.